

Electrical Engineering BSc

Electric Machinery II

Exercises and Industrial Application Problems

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Preface

This collection of exercises and industrial application problems is organised into three chapters:

- DC machines
- Three-phase synchronous machines
- Electrical machine transients

This document presents exercises of a more conceptual nature alongside industrial application problems, allowing students to progress from fundamental concepts to practical situations encountered in industry. Each exercise or problem is numbered sequentially (Exercise 1, Exercise 2, ...), with its scope made clear by the context and problem statement itself.

The odd-numbered exercises are fully worked through, while the even-numbered exercises are proposed as work items, with only the solutions given. The latter are primarily intended to be solved during theoretical-practical classes, adopting Problem-Based Learning as the teaching and learning methodology. In this way, discussion and critical analysis of the topics are encouraged, fostering the development of analytical thinking and problem-solving skills.

Among the worked exercises, some are solved analytically as an introduction to, or consolidation of, key concepts, while others are solved using the scientific computing language *Julia*, through computational documents (*notebooks*) developed using the integrated development environment *Pluto.jl*. The use of interactive *notebooks* allows students to deepen their understanding of fundamental theoretical concepts and to develop essential skills for professional practice. In addition, this exploratory practice allows students to apply their knowledge to solve exercises interactively, to visualise and analyse results graphically, and to gain practical experience with engineering tools.

Lisbon, September 2025

Ricardo Luís

*The real process of education should be
the process of learning to think through
the application of real problems.*

— **John Dewey** (1859-1952)

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DC MACHINES

Exercise 1

A six-pole DC machine has 360 conductors in armature slots. Each magnetic pole has a pole arc of 20 cm, a depth of 20 cm and a magnetic flux density of 0.8 T. With the rotor running at a speed of 1000 rpm, determine the induced electromotive force, E_0 , obtained if the machine has:

- a lap-wound armature winding;
- a wave-wound armature winding.

Worked solution:

Let:

- $2p = 6$ poles; $z = 360$ conductors;
- $b = 20$ cm (pole arc); $l = 20$ cm (depth)
- $B = 0.8$ T

Pole face area, A_p :

$$A_p = b l = 0.2 \times 0.2 = 0.04 \text{ m}^2$$

The magnetic flux per pole, ϕ_p , is then:

$$\phi_p = B A_p = 0.8 \times 0.04 = 0.032 \text{ Wb}$$

With the speed in rpm, the armature constant, k , is given by: $k = \frac{z p}{60 a}$

Thus, the induced electromotive force, E_0 , in the armature winding is obtained:

a) Lap winding, $a = p$ (number of brushes equal to number of poles):

$$E_0 = k \phi_p n = \frac{360 \times 3}{60 \times 3} \times 0.032 \times 1000 = 192 \text{ V}$$

b) Wave winding, $a = 1$ (only one pair of brushes):

$$E_0 = k \phi_p n = \frac{360 \times 3}{60 \times 1} \times 0.032 \times 1000 = 576 \text{ V}$$

(Source: Adapted from exercise 5.5 in [Guru & Hızıroğlu, 2003])

Exercise 2

Consider the armature of a four-pole DC generator with a lap winding placed in 28 slots with 10 conductors in each. The magnetic flux per pole is 40 mWb and the armature (rotor) speed is 1200 rpm. The generator supplies a load, and the current in each conductor is 2 A. What are the torque and power developed by the generator?

Answers:

$$T_d = 14.26 \text{ Nm} \quad ; \quad P_d = 1.79 \text{ kW}$$

(Source: Adapted from exercise 5.8 in [Guru & Hızıroğlu, 2003])

Exercise 3

Consider a DC generator with $2p = 6$ poles, a simple lap armature winding, i.e. $2a = 6$ parallel paths, and $z = 624$ active conductors uniformly distributed around the armature periphery.

The cross-sectional area of each armature conductor is $s = 8 \text{ mm}^2$, and carries a current density of $J = 3 \text{ A.mm}^{-2}$.

The machine has a pole-arc-to-pole pitch ratio, $b/\tau = 2/3$, and the armature has a diameter of $D = 220 \text{ mm}$.

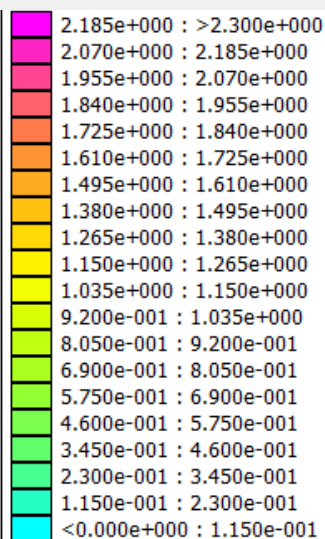
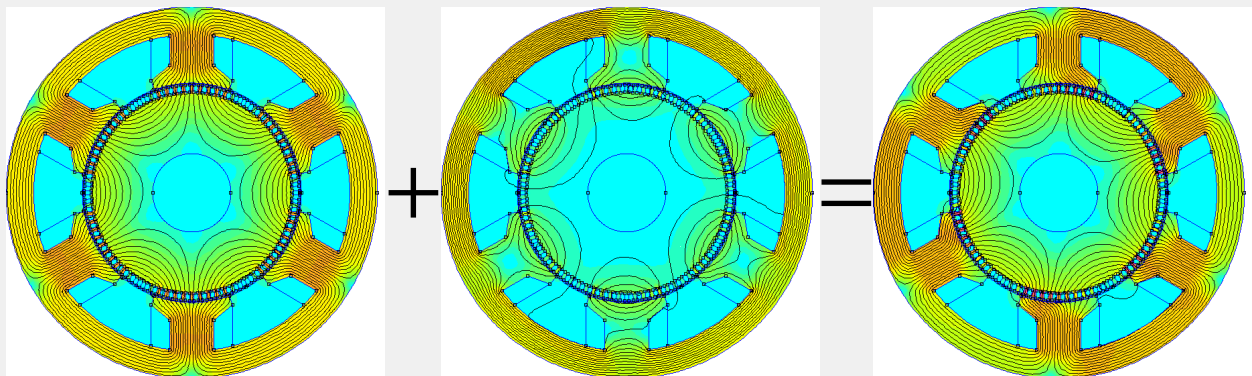
- a) Determine the magnetomotive force (MMF) associated with the armature reaction, $\mathcal{F}_i$, along the geometric neutral axis;
- b) Assuming that the machine is fitted with interpoles only, determine the number of turns to be placed on the interpoles;
- c) Suppose the machine is to have its $\mathcal{F}_i$ fully compensated:
 - 1) Calculate the number of turns to be placed in a compensating winding, assuming that 6 slots were opened in each main pole. Calculate the number of conductors to be placed in each slot;
 - 2) Size the number of turns to be placed on the interpoles after the compensating windings have been installed.

Worked solution:

Two solution approaches were used in this exercise:

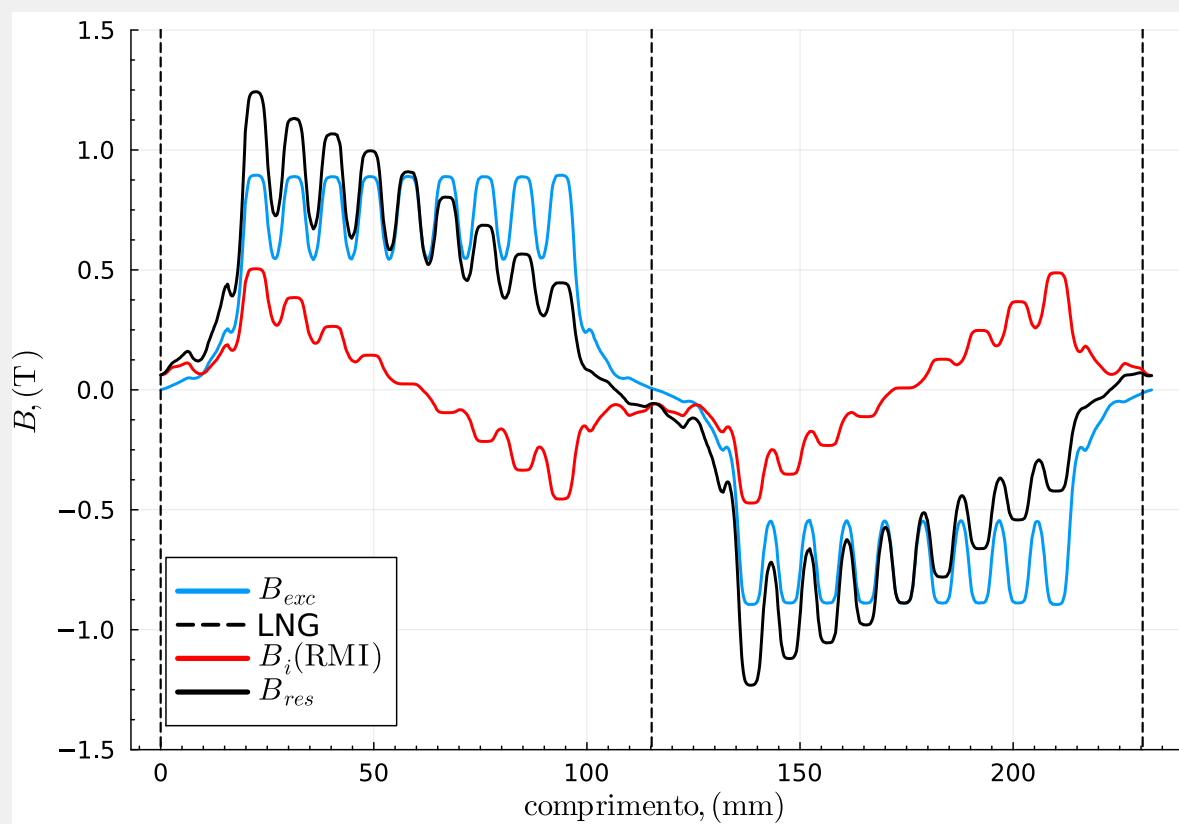
- **numerical (computational) solution:** using the exercise data, a CAD (*Computer-Aided Design*) and CAE (*Computer-Aided Engineering*) model was built, applying the finite element method for the electromagnetic calculation. The *software* used was FEMM - Finite Element Method Magnetics. The computational solutions make it possible to visualise and quantify the magnetic flux lines, helping to understand the armature magnetic reaction and its effects on the machine's operation;
- **Analytical solution:** this is the main objective of this exercise, namely to compensate the Armature Magnetic Reaction (AMR), by calculating the magnetomotive forces involved.

field winding + armature field = resultant field

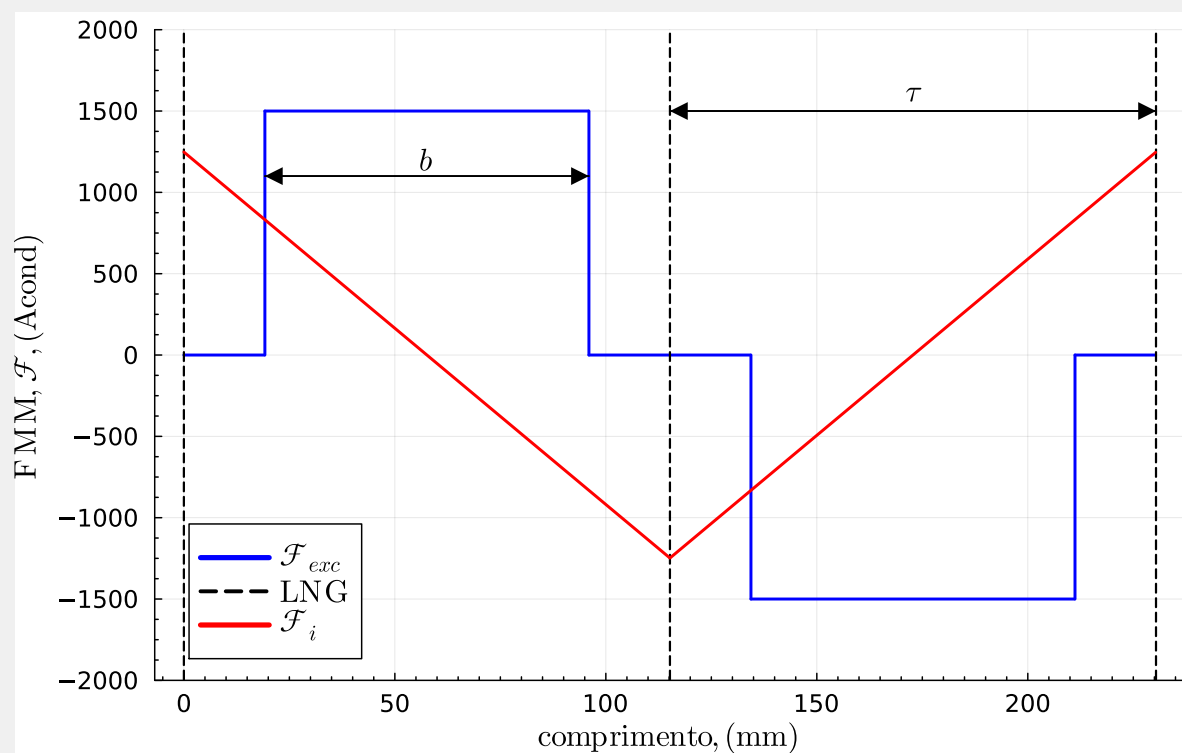


Density Plot: |B|, Tesla

field winding + armature field = resultant field



MMF of the field winding and the armature reaction



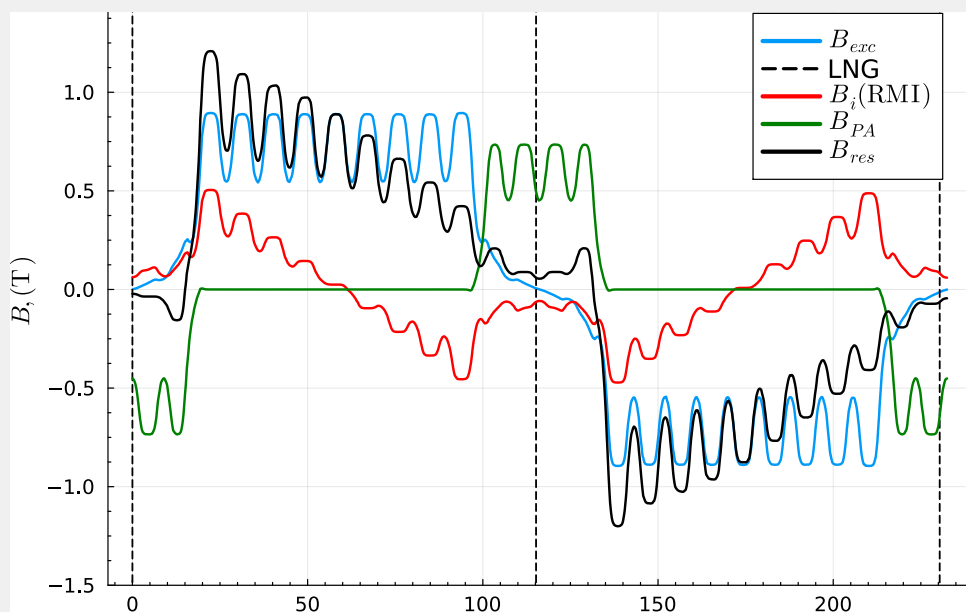
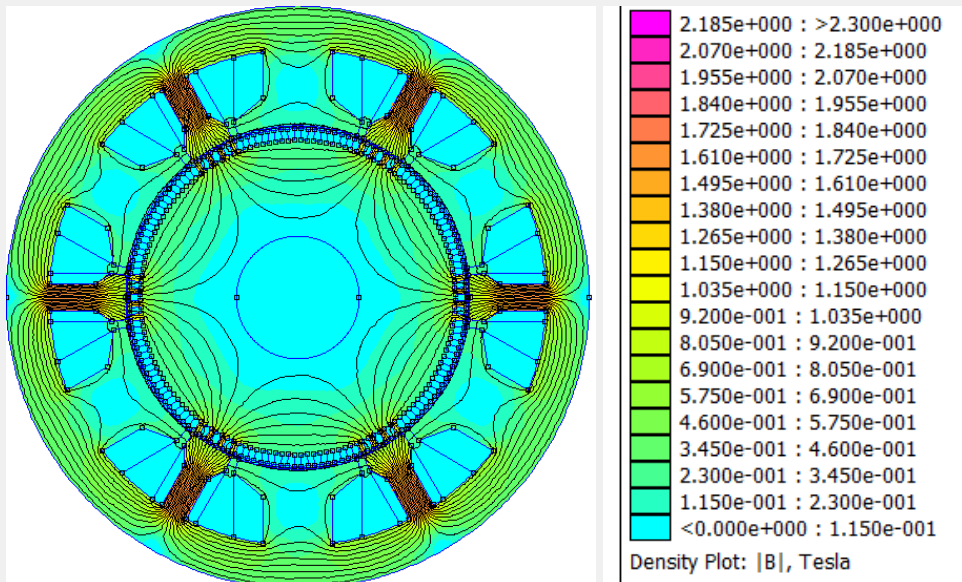
a)

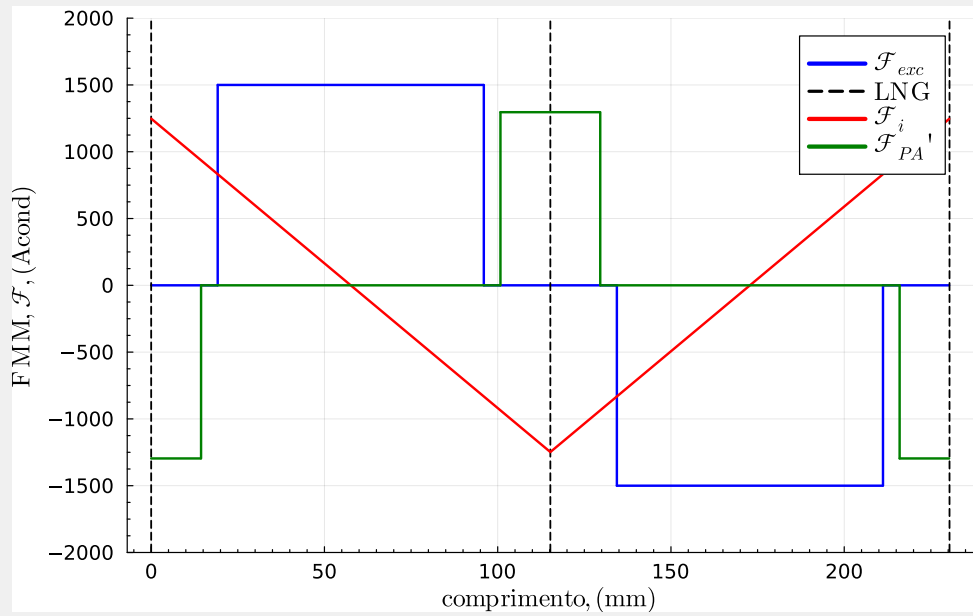
$$I_a = J s = 3 \times 8 = 24 \text{ A}$$

$$I = I_a 2a = 24 \times 2 \times 3 = 144 \text{ A}$$

$$\mathcal{F}_i = A \frac{\tau}{2} \quad \text{with:} \quad A = \frac{z I}{4 a p} \frac{1}{\tau} \quad \text{or} \quad \mathcal{F}_i = \frac{z I}{8 a p}, \quad \text{using this latter expression:}$$

$$\mathcal{F}_i = \frac{z I}{8 a p} = \frac{624 \times 144}{8 \times 3 \times 3} = 1248 \text{ A.cond}$$

b) DC machine with **interpoles**:



For a machine with interpoles:

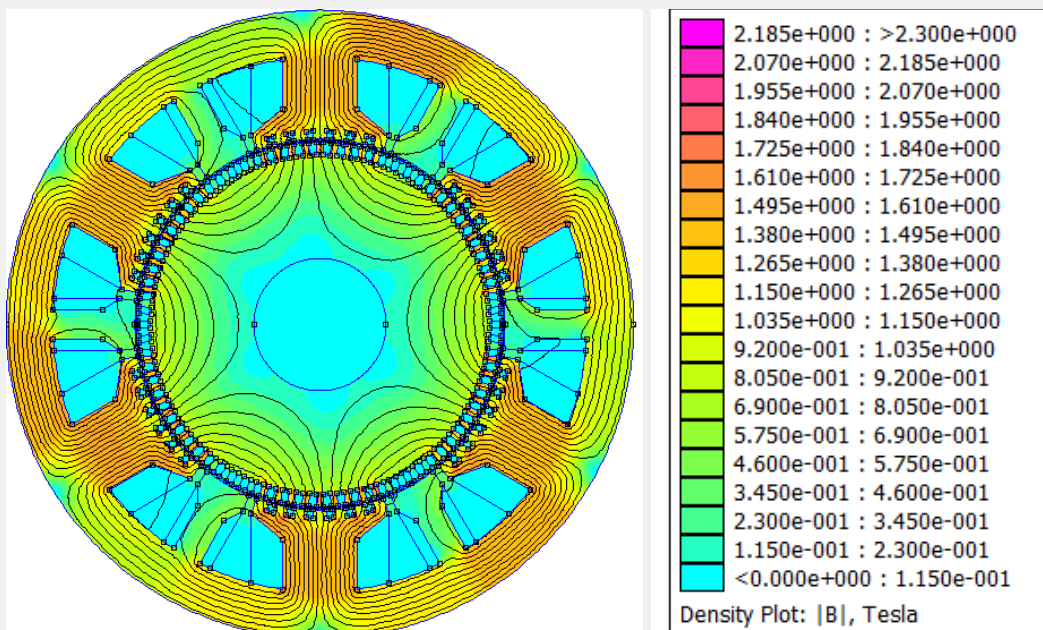
$$\mathcal{F}_{PA} = \mathcal{F}_i \text{ since: } \mathcal{F}_{PA} = N_{PA} I, \text{ we obtain: } N_{PA} = \frac{\mathcal{F}_i}{I} = \frac{1248}{144} = 8.67 \text{ turns}$$

To facilitate commutation, the next integer above is used, i.e. 9 turns.

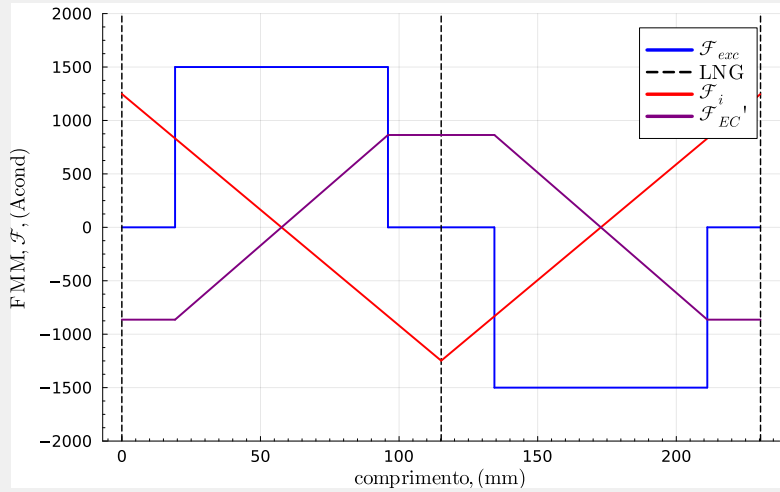
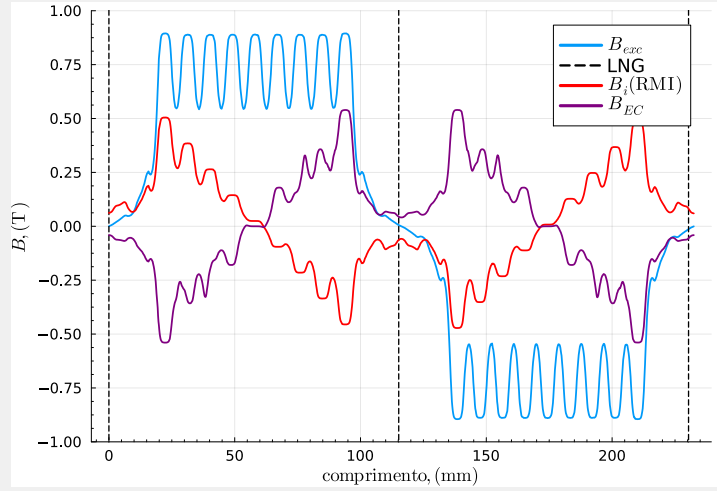
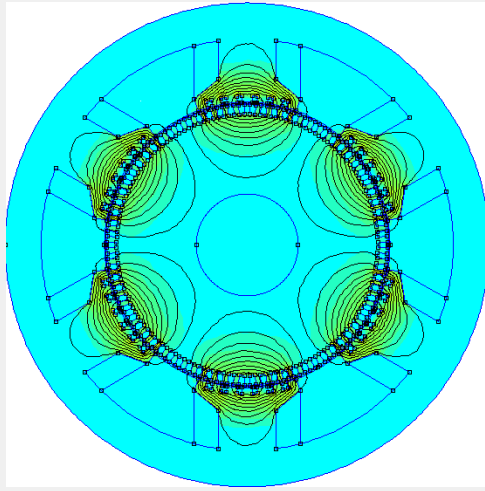
Thus, a corrected MMF is obtained for the interpoles, $\mathcal{F}'_{PA}$:

$$\mathcal{F}'_{PA} = 9 \times 144 = 1296 \text{ A.cond}$$

c) Machine with compensated AMR (interpoles and compensating windings)



c-1) The compensating windings are sized first:



$\mathcal{F}_{EC} = A \frac{b}{2}$ or $\mathcal{F}_{EC} = \mathcal{F}_i \frac{b}{\tau}$, using this latter expression since $\frac{b}{\tau}$ is known:

$$\mathcal{F}_{EC} = \mathcal{F}_i \frac{b}{\tau} = 1248 \times \frac{2}{3} = 832 \text{ A.cond}$$

On the other hand, since $\mathcal{F}_{EC} = N_{EC} I$, the number of turns of the compensating winding is obtained as:

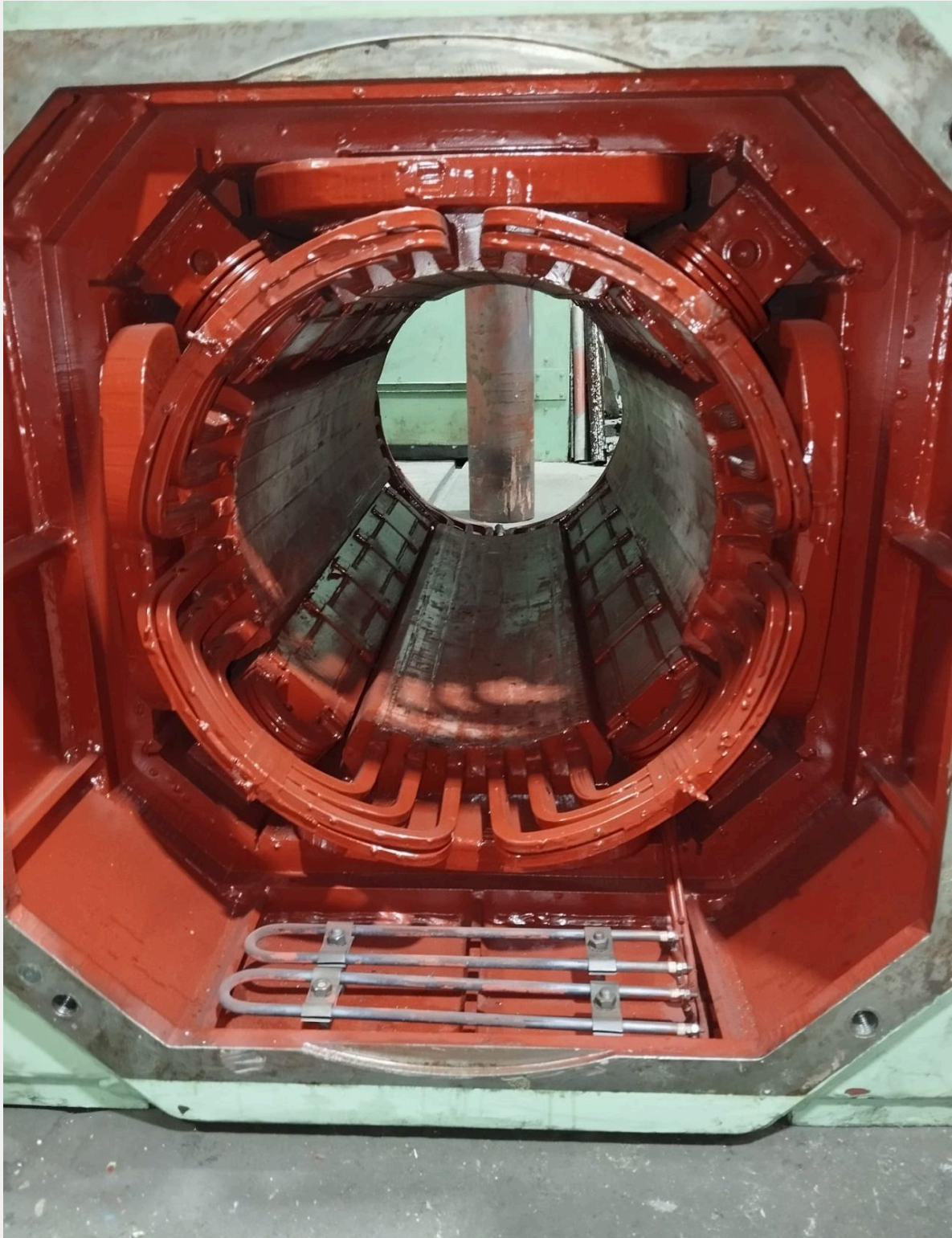
$$\text{Thus, } N_{EC} = \frac{\mathcal{F}_{EC}}{I} = \frac{832}{144} = 5.78 \text{ turns}$$

Considering 6 slots per main pole, where the turns from 3 slots connect to the following pole and those from the other 3 slots connect to the preceding pole, $N'_{EC} = 6$ turns are obtained, placing 2 turns in each slot.

Thus, a corrected MMF is obtained for the compensating windings, $\mathcal{F}'_{EC}$:

$$\mathcal{F}'_{EC} = 6 \times 144 = 864 \text{ A.cond}$$

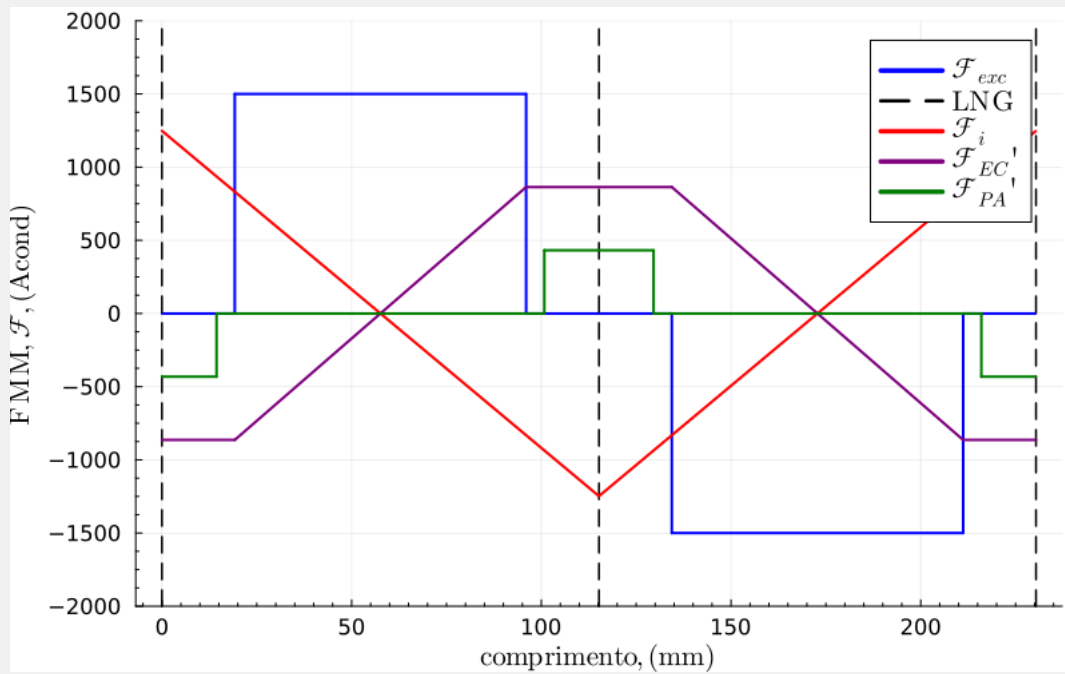
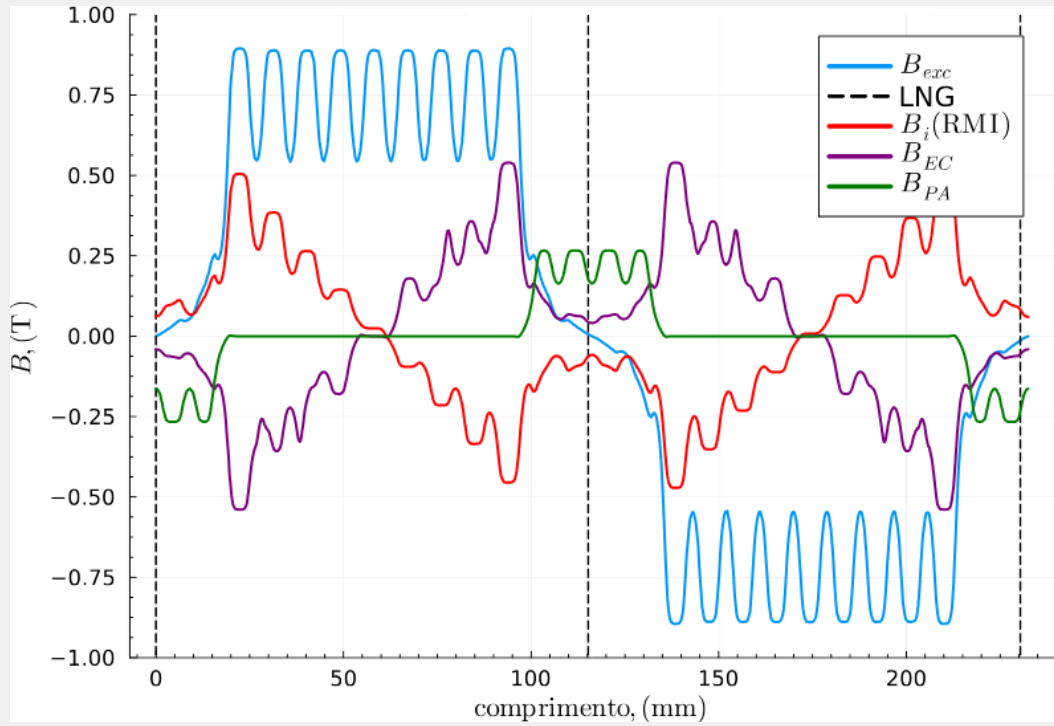
Example photograph of a machine with compensating windings (and interpoles) with 6 slots in the pole-shoe head and 2 turns per slot:



Photograph from refurbishment work on a DC machine.

Image of the stator with interpoles and compensating windings (source: [Lusonave](#)).

c-2) and then the **interpoles**:



$$\mathcal{F}_{PA} = \mathcal{F}_i - \mathcal{F}'_{EC} = 1248 - 864 = 384 \text{ A.cond}$$

$$N_{PA} = \frac{\mathcal{F}_{PA}}{I} = \frac{384}{144} = 2.67 \text{ cond}$$

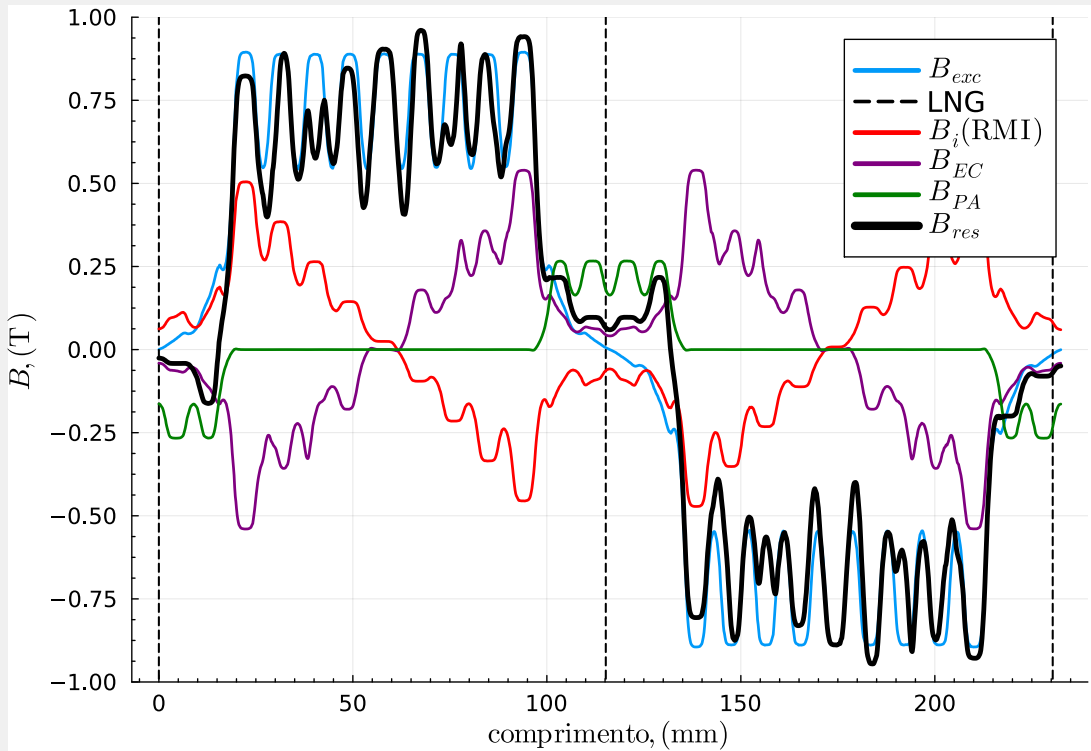
To facilitate commutation, the next integer above is used, i.e. 3 turns.

Thus, a corrected MMF is obtained for the interpoles, $\mathcal{F}'_{PA}$:

$$\mathcal{F}'_{PA} = 3 \times 144 = 432 \text{ A.cond}$$

Conclusion: after fitting the compensating windings and the interpoles, a resultant MMF, $\mathcal{F}_R$, is obtained at the Geometric Neutral Lines (brush positions):

$$\mathcal{F}_R = \mathcal{F}_i - \mathcal{F}'_{PEC} - \mathcal{F}'_{PA} = 1248 - 864 - 432 = -48 \text{ A.cond}$$



Exercise 4

An armature winding of a dynamo has $z = 640$ conductors, wound as a simple lap winding, for a rated current of 100 A and 2 pole pairs.

- Size the number of turns of the compensating windings, N_{EC} , and of the interpoles, N_{PA} , for full compensation of the armature magnetic reaction. Consider that the machine's field poles cover 70 % of the total periphery, as seen from the rotor;
- Resize N_{EC} and N_{PA} considering a simple wave armature winding.

Answers:

- a) $N_{EC} = 14$ turns; $N_{PA} = 6$ turns
b) $N_{EC} = 28$ turns; $N_{PA} = 12$ turns